

Table of Contents

Uralstech.AvLoader	3
AvFlow	5
AvFlow.Builder	8
AvGender	13
AvImageFileExtension	14
AvMetadata	15
AvModelFileExtension	20
AvSourceData	21
AvType	24
IAsyncAvPostProcessor	25
IAvDataLoader	26
IAvImporter	27
IAvPostProcessor	29
LoadedAv	30
Uralstech.AvLoader.Capabilities	38
BlendShapeProviderExtensions	39
IAvatarExpressionProvider	40
IBlendShapeProvider	41
ICapability	44
IImportedAnimator	45
IMeshBlendShapeProvider	46
Uralstech.AvLoader.DataLoaders	47
FileAvDataLoader	48
URIAvDataLoader	56
Uralstech.AvLoader.Importers	60
GLTFastAvImporter	61
LoadedGLTFastAv	65
LoadedUniGLTFAv	70
LoadedUniVRM10Av	75
UniGLTFAvImporter	81
UniVRM10AvImporter	85
Uralstech.AvLoader.PostProcessors	90
ActionPostProcessor	91
AnimatorPostProcessor	93
AsyncFuncPostProcessor	96
AvPost	98
CachePostProcessor	107
ConditionalPostProcessor	111

IAvatarDependentComponent	114
MaterialOverrideConfig	115
MaterialOverridePostProcessor	116
ShaderSwapConfig	118
ShaderSwapPostProcessor	119
Uralstech.AvLoader.PostProcessors.Rendering	121
RuntimeMaterialOverrideDefinition	122
ShaderKeywordMapping	125
ShaderKeywordRule	127
ShaderMapping	130
ShaderPropertyMapping	132
ShaderPropertyType	134
Uralstech.AvLoader.Utils	135
ListExtensions	136
TextureExtensions	137
UnityWebRequestException	140
WebRequestExtensions	142

Namespace Uralstech.AvLoader

Classes

[AvFlow](#)

Class supporting a complete avatar loader flow with fallbacks for each step.

[AvFlow.Builder](#)

Fluent builder for [AvFlow](#).

[AvMetadata](#)

Information regarding a loaded model.

[AvSourceData](#)

Wrapper for a raw container of avatar data.

[LoadedAv](#)

A loaded avatar and its associated data.

Interfaces

[IAsyncAvPostProcessor](#)

A simple script that can perform an asynchronous post-processing step on an imported avatar.

[IAvDataLoader](#)

Interface for an avatar data loader.

[IAvImporter](#)

Interface for an avatar importer.

[IAvPostProcessor](#)

A simple script that can perform a post-processing step on an imported avatar.

Enums

[AvGender](#)

The gender a loaded avatar or of the outfit it is wearing. Can be useful for using specific animation avatars for different types.

[AvImageFileExtension](#)

The file extension of an avatar's render(s).

[AvModelFileExtension](#)

The file extension of an avatar's model.

[AvType](#)

Describes the type of a loaded model.

Class AvFlow

Namespace: [Uralstech.AvLoader](#)

Class supporting a complete avatar loader flow with fallbacks for each step.

```
public class AvFlow
```

Inheritance

object ← AvFlow

Constructors

AvFlow(IReadOnlyList<IAvDataLoader>, IReadOnlyList<IAvImporter>)

```
public AvFlow(IReadOnlyList<IAvDataLoader> dataLoaders, IReadOnlyList<IAvImporter> importers)
```

Parameters

dataLoaders IReadOnlyList<[IAvDataLoader](#)>

importers IReadOnlyList<[IAvImporter](#)>

Fields

AsyncPostProcessors

Async post-processors to run after loading the avatar.

```
public IEnumerable<IAsyncAvPostProcessor> AsyncPostProcessors
```

Field Value

IEnumerable<[IAsyncAvPostProcessor](#)>

DataLoaders

The loaders to use in the flow, in priority order.

```
public readonly IReadOnlyList<IAvDataLoader> DataLoaders
```

Field Value

IReadOnlyList<[IAvDataLoader](#)>

Remarks

Index 0 is used first; if it cannot handle the data, the flow falls back to index 1, then 2, and so on.

Importers

The importers to use in the flow, in priority order.

```
public readonly IReadOnlyList<IAvImporter> Importers
```

Field Value

IReadOnlyList<[IAvImporter](#)>

Remarks

Index 0 is used first; if it cannot handle the model format, the flow falls back to index 1, then 2, and so on.

PostProcessors

Post-processors to run after loading the avatar.

```
public IEnumerable<IAvPostProcessor> PostProcessors
```

Field Value

IEnumerable<[IAvPostProcessor](#)>

Methods

RunAsync(CancellationToken)

Runs the loader flow completely.

```
public Awaitable<LoadedAv> RunAsync(CancellationToken token = default)
```

Parameters

token CancellationToken

Returns

Awaitable<[LoadedAv](#)>

The loaded avatar.

TryRunAsync(CancellationToken)

Tries to run the loader flow completely.

```
public Awaitable<LoadedAv?> TryRunAsync(CancellationToken token = default)
```

Parameters

token CancellationToken

Returns

Awaitable<[LoadedAv](#)>

The loaded avatar if the entire flow completed successfully; [null](#) otherwise.

Class AvFlow.Builder

Namespace: [Uralstech.AvLoader](#)

Fluent builder for [AvFlow](#).

```
public sealed class AvFlow.Builder
```

Inheritance

object ← AvFlow.Builder

Methods

Build()

Builds the [AvFlow](#).

```
public AvFlow Build()
```

Returns

[AvFlow](#)

Remarks

The returned object is independent of the builder and will not reflect subsequent changes made to this builder instance.

Copy()

Creates a copy of the builder.

```
public AvFlow.Builder Copy()
```

Returns

[AvFlow.Builder](#)

Remarks

The returned object is independent of the builder and will not reflect subsequent changes made to this builder instance.

From(IAvDataLoader)

Adds an avatar data loader to the flow.

```
public AvFlow.Builder From(IAvDataLoader loader)
```

Parameters

loader [IAvDataLoader](#)

Returns

[AvFlow.Builder](#)

Remarks

Loaders are tried in the order they are added until one succeeds.

From(params IAvDataLoader[])

Adds multiple avatar data loaders to the flow.

```
public AvFlow.Builder From(params IAvDataLoader[] loaders)
```

Parameters

loaders [IAvDataLoader\[\]](#)

Returns

[AvFlow.Builder](#)

Remarks

Loaders are tried in the order they are added until one succeeds.

ProcessWith(IAvPostProcessor)

Adds a post-processor to the flow.

```
public AvFlow.Builder ProcessWith(IAvPostProcessor postProcessor)
```

Parameters

postProcessor [IAvPostProcessor](#)

Returns

[AvFlow.Builder](#)

Remarks

Note that **all** sync post-processors run *before* **all** async post-processors, regardless of order of addition in the builder.

ProcessWith(params IAvPostProcessor[])

Adds multiple post-processors to the flow.

```
public AvFlow.Builder ProcessWith(params IAvPostProcessor[] postProcessors)
```

Parameters

postProcessors [IAvPostProcessor\[\]](#)

Returns

[AvFlow.Builder](#)

Remarks

Note that **all** sync post-processors run *before* **all** async post-processors, regardless of order of addition in the builder.

ProcessWithAsync(IAsyncAvPostProcessor)

Adds an asynchronous post-processor to the flow.

```
public AvFlow.Builder ProcessWithAsync(IAsyncAvPostProcessor asyncPostProcessor)
```

Parameters

asyncPostProcessor [IAsyncAvPostProcessor](#)

Returns

[AvFlow.Builder](#)

Remarks

Note that **all** async post-processors run *after* **all** sync post-processors, regardless of order of addition in the builder.

ProcessWithAsync(params IAsyncAvPostProcessor[])

Adds multiple asynchronous post-processors to the flow.

```
public AvFlow.Builder ProcessWithAsync(params IAsyncAvPostProcessor[] asyncPostProcessors)
```

Parameters

asyncPostProcessors [IAsyncAvPostProcessor\[\]](#)

Returns

[AvFlow.Builder](#)

Remarks

Note that **all** async post-processors run *after* **all** sync post-processors, regardless of order of addition in the builder.

Using(IAvImporter)

Adds an avatar importer to the flow.

```
public AvFlow.Builder Using(IAvImporter importer)
```

Parameters

importer [IAvImporter](#)

Returns

[AvFlow.Builder](#)

Remarks

Importers are evaluated based on format support, then tried in the order they are added until one succeeds.

Using(params IAvImporter[])

Adds multiple avatar importers to the flow.

```
public AvFlow.Builder Using(params IAvImporter[] importers)
```

Parameters

importers [IAvImporter\[\]](#)

Returns

[AvFlow.Builder](#)

Remarks

Importers are evaluated based on format support, then tried in the order they are added until one succeeds.

Enum AvGender

Namespace: [Uralstech.AvLoader](#)

The gender a loaded avatar or of the outfit it is wearing. Can be useful for using specific animation avatars for different types.

```
public enum AvGender
```

Fields

```
[EnumMember(Value = "feminine")] Feminine = 2
```

```
[EnumMember(Value = "masculine")] Masculine = 1
```

```
[EnumMember(Value = "neutral")] Neutral = 3
```

```
None = 0
```

Enum AvImageFileExtension

Namespace: [Uralstech.AvLoader](#)

The file extension of an avatar's render(s).

```
public enum AvImageFileExtension
```

Fields

JPEG = 1

JPEGAny = 3

JPG = 2

None = 0

PNG = 4

Class AvMetadata

Namespace: [Uralstech.AvLoader](#)

Information regarding a loaded model.

```
public class AvMetadata
```

Inheritance

object ← AvMetadata

Constructors

AvMetadata(string)

```
public AvMetadata(string id)
```

Parameters

id string

Fields

AdditionalMetadata

Optional additional metadata for the avatar.

```
public Dictionary<string, string>? AdditionalMetadata
```

Field Value

Dictionary<string, string>

CreatedAt

The creation timestamp of the model.

```
public DateTimeOffset CreatedAt
```

Field Value

DateTimeOffset

Gender

The gender of the model or outfit it is wearing.

```
public AvGender Gender
```

Field Value

[AvGender](#)

Id

Unique ID of the avatar.

```
public string Id
```

Field Value

string

SkinTone

The skin tone of the model stored as a hex string (including the #).

```
public string? SkinTone
```

Field Value

string

Type

The type of the model.

```
public AvType Type
```

Field Value

[AvType](#)

UpdatedAt

The last time this model was updated.

```
public DateTimeOffset UpdatedAt
```

Field Value

DateTimeOffset

Methods

TryCreateFromBytes(ReadOnlySpan<byte>, out AvMetadata?, Encoding?, bool)

Tries creating a [AvMetadata](#) struct from raw bytes.

```
public static bool TryCreateFromBytes(ReadOnlySpan<byte> data, out AvMetadata? metadata,
    Encoding? encoding = null, bool throwOnFail = false)
```

Parameters

data ReadOnlySpan<byte>

The data to read from.

metadata [AvMetadata](#)

The decoded struct, [null](#) if decoding fails.

encoding Encoding

The encoding of the data. Assumes System.Text.Encoding.UTF8 if [null](#).

throwOnFail bool

Should the method throw if it couldn't create the [AvMetadata](#)?

Returns

bool

[true](#) if successful; [false](#) otherwise.

TryCreateFromFile(string, out AvMetadata?, bool)

Tries creating a [AvMetadata](#) struct from a file.

```
public static bool TryCreateFromFile(string path, out AvMetadata? metadata, bool throwOnFail = false)
```

Parameters

path string

The path to read from.

metadata [AvMetadata](#)

The decoded struct, [null](#) if decoding fails.

throwOnFail bool

Should the method throw if it couldn't create the [AvMetadata](#)?

Returns

bool

[true](#) if successful; [false](#) otherwise.

TryCreateFromUriAsync(Uri, CancellationToken?, Encoding?, bool)

Tries creating a [AvMetadata](#) struct from data at the given URI.

```
public static Awaitable<AvMetadata?> TryCreateFromUriAsync(Uri uri, CancellationToken? token = null, Encoding? encoding = null, bool throwOnFail = false)
```

Parameters

uri Uri

The URI to load from.

token CancellationToken?

encoding Encoding

The encoding of the data. Assumes System.Text.Encoding.UTF8 if [null](#).

throwOnFail bool

Should the method throw if it couldn't create the [AvMetadata](#)?

Returns

Awaitable<[AvMetadata](#)>

The decoded [AvMetadata](#) struct if successful; [null](#) otherwise.

Enum AvModelFileExtension

Namespace: [Uralstech.AvLoader](#)

The file extension of an avatar's model.

```
public enum AvModelFileExtension
```

Fields

GLB = 2

GLTF = 1

GLTFAny = 3

None = 0

VRM = 4

Class AvSourceData

Namespace: [Uralstech.AvLoader](#)

Wrapper for a raw container of avatar data.

```
public class AvSourceData
```

Inheritance

object ← AvSourceData

Constructors

AvSourceData(byte[], AvModelFileExtension, string?, AvMetadata, Type, Texture2D?, Texture2D?)

```
public AvSourceData(byte[] model, AvModelFileExtension modelFormat, string? modelPath,
    AvMetadata metadata, Type dataLoaderType, Texture2D? fullRender = null, Texture2D?
    bustRender = null)
```

Parameters

model byte[]

modelFormat [AvModelFileExtension](#)

modelPath string

metadata [AvMetadata](#)

dataLoaderType Type

fullRender Texture2D?

bustRender Texture2D?

Fields

BustRender

Optional bust render of the avatar.

```
public readonly Texture2D? BustRender
```

Field Value

Texture2D?

DataLoaderType

The type of the data loader which created this.

```
public readonly Type DataLoaderType
```

Field Value

Type

FullRender

Optional full render of the avatar.

```
public readonly Texture2D? FullRender
```

Field Value

Texture2D?

Metadata

Metadata of the avatar.

```
public readonly AvMetadata Metadata
```

Field Value

[AvMetadata](#)

Model

The model as a byte[].

```
public readonly byte[] Model
```

Field Value

byte[]

ModelFormat

The file format of the model.

```
public readonly AvModelFileExtension ModelFormat
```

Field Value

[AvModelFileExtension](#)

ModelPath

The path/URI to model.

```
public readonly string? ModelPath
```

Field Value

string

Enum AvType

Namespace: [Uralstech.AvLoader](#)

Describes the type of a loaded model.

```
public enum AvType
```

Fields

```
[EnumMember(Value = "fullbody")] HumanoidFullBody = 1
```

A humanoid, full-body model.

```
[EnumMember(Value = "fullbody-xr")] HumanoidFullBodyXR = 3
```

A humanoid, full-body model for XR use.

```
[EnumMember(Value = "halfbody")] HumanoidHalfBody = 2
```

A humanoid, half-body model.

```
None = 0
```


Interface IAsyncAvPostProcessor

Namespace: [Uralstech.AvLoader](#)

A simple script that can perform an asynchronous post-processing step on an imported avatar.

```
public interface IAsyncAvPostProcessor
```

Methods

PostProcessAsync(LoadedAv, AvSourceData, CancellationToken)

Runs a post-processing step on the given avatar.

```
Awaitable PostProcessAsync(LoadedAv avatar, AvSourceData rawData, CancellationToken token  
= default)
```

Parameters

avatar [LoadedAv](#)

The loaded avatar to process.

rawData [AvSourceData](#)

The raw data of the loaded avatar.

token [CancellationToken](#)

Returns

Awaitable

Interface IAvDataLoader

Namespace: [Uralstech.AvLoader](#)

Interface for an avatar data loader.

```
public interface IAvDataLoader
```

Remarks

The role of the [IAvDataLoader](#) is to fetch the raw binary and JSON data associated with the avatar. They can be chained up to create fallbacks, for example, a [FileAvDataLoader](#) could check local caches for existing avatars, then fall back to a [URIAvDataLoader](#) to download it from remote sources.

You could also implement your own data loaders to load avatars using APIs provided by cloud services, like Firebase.

Methods

LoadAvatarAsync(bool, CancellationToken)

Tries loading an avatar into an [AvSourceData](#).

```
Awaitable<AvSourceData?> LoadAvatarAsync(bool throwOnFail, CancellationToken token  
= default)
```

Parameters

throwOnFail bool

Should this method throw errors on failures or log them as warnings and return [null](#)?

token CancellationToken

Returns

Awaitable<[AvSourceData](#)>

The loaded data if successful; [null](#) on failure.

Interface IAvImporter

Namespace: [Uralstech.AvLoader](#)

Interface for an avatar importer.

```
public interface IAvImporter
```

Remarks

The role of the [IAvImporter](#) is to parse data returned by an [IAvDataLoader](#) and bring the avatar into Unity-space as a GameObject. Like [IAvDataLoader](#)s, they can be chained up to create fallbacks, for example, a [GLTFastAvImporter](#) for glTF support, falling back to one for .fbx support, etc.

Methods

ImportAvatarAsync(AvSourceData, bool, CancellationToken)

Tries to import the avatar into a scene as a disabled GameObject, along with metadata and any renders.

```
Awaitable<LoadedAv?> ImportAvatarAsync(AvSourceData rawData, bool throwOnFail,  
CancellationToken token = default)
```

Parameters

rawData [AvSourceData](#)

The raw avatar data to process.

throwOnFail bool

Should this method throw errors on failures or log them as warnings and return [null](#)?

token CancellationToken

Returns

Awaitable<[LoadedAv](#)>

The loaded avatar if successful; [null](#) on failure.

Remarks

The returned [LoadedAv](#) may contain code to handle requirements of format-supporting plugins. For example, [GLTFastAvImporter](#) depends on glTFast, which requires that GltfImport object used to import the avatar be active alongside the avatar's GameObject, and should be disposed after the avatar is no longer needed. Thus, [LoadedAv](#) implements System.IDisposable.

SupportsFormat(AvModelFileExtension)

Returns [true](#) if this importer can handle the given format; [false](#) otherwise.

```
bool SupportsFormat(AvModelFileExtension format)
```

Parameters

format [AvModelFileExtension](#)

Returns

bool

Interface IAvPostProcessor

Namespace: [Uralstech.AvLoader](#)

A simple script that can perform a post-processing step on an imported avatar.

```
public interface IAvPostProcessor
```

Methods

PostProcess(LoadedAv, AvSourceData)

Runs a post-processing step on the given avatar.

```
void PostProcess(LoadedAv avatar, AvSourceData rawData)
```

Parameters

avatar [LoadedAv](#)

The loaded avatar to process.

rawData [AvSourceData](#)

The raw data of the loaded avatar.

Class LoadedAv

Namespace: [Uralstech.AvLoader](#)

A loaded avatar and its associated data.

```
public abstract class LoadedAv
```

Inheritance

object ← LoadedAv

Derived

[LoadedGLTFastAv](#), [LoadedUniGLTFAv](#), [LoadedUniVRM10Av](#)

Constructors

LoadedAv(GameObject, AvMetadata, Texture2D?, Texture2D?, Type)

```
protected LoadedAv(GameObject gameObject, AvMetadata metadata, Texture2D? fullRender, Texture2D? bustRender, Type importerType)
```

Parameters

gameObject GameObject

metadata [AvMetadata](#)

fullRender Texture2D?

bustRender Texture2D?

importerType Type

Fields

BustRender

Optional bust render of the avatar.

```
public readonly Texture2D? BustRender
```

Field Value

Texture2D?

DestroyOnDispose

Deprecated. Use [LifetimeObjects](#) and the [RegisterLifetimeObject\(Object\)](#) APIs instead.

```
[Obsolete("Use LifetimeObjects/RegisterLifetimeObject/DeregisterLifetimeObject instead.")]  
public readonly List<UnityEngine.Object> DestroyOnDispose
```

Field Value

List<UnityEngine.Object>

FullRender

Optional full render of the avatar.

```
public readonly Texture2D? FullRender
```

Field Value

Texture2D?

GameObject

The avatar's GameObject.

```
public readonly GameObject GameObject
```

Field Value

GameObject

ImporterType

The type of the importer which created this [LoadedAv](#).

```
public readonly Type ImporterType
```

Field Value

Type

Metadata

Metadata of the avatar.

```
public readonly AvMetadata Metadata
```

Field Value

[AvMetadata](#)

Properties

LifetimeObjects

Unity objects whose lifetime is owned by this avatar.

```
public IReadOnlyCollection<UnityEngine.Object> LifetimeObjects { get; }
```

Property Value

IReadOnlyCollection<UnityEngine.Object>

Remarks

All registered objects are destroyed when the avatar is disposed. [FullRender](#) and [BustRender](#) are registered automatically. Call [DeregisterLifetimeObject\(Object\)](#) to take ownership of an object and prevent it from being destroyed.

Methods

DeregisterLifetimeObject(Object)

Removes a Unity object from the avatar's lifetime ownership.

```
public void DeregisterLifetimeObject(UnityEngine.Object unityObject)
```

Parameters

unityObject Object

Remarks

After deregistration, the caller becomes responsible for destroying the object. This is can be used to retain [FullRender](#) or [BustRender](#).

Dispose()

Performs application-defined tasks associated with freeing, releasing, or resetting unmanaged resources.

```
public void Dispose()
```

GetAvatarMaterials(bool)

Gets all materials used by the avatar.

```
public IReadOnlyList<Material> GetAvatarMaterials(bool refreshCache = false)
```

Parameters

refreshCache bool

Returns

IReadOnlyList<Material>

Remarks

Results are cached. Call with **refreshCache** set to [true](#) if renderers or materials change at runtime.

GetAvatarMeshes(bool)

Gets all meshes used by the avatar.

```
public IReadOnlyList<Mesh> GetAvatarMeshes(bool refreshCache = false)
```

Parameters

refreshCache bool

Returns

IReadOnlyList<Mesh>

Remarks

Results are cached. Call with **refreshCache** set to [true](#) if renderers or meshes change at runtime.

GetAvatarRenderers(bool)

Gets all renderers used by the avatar.

```
public IReadOnlyList<Renderer> GetAvatarRenderers(bool refreshCache = false)
```

Parameters

refreshCache bool

Returns

ReadOnlyList<Renderer>

Remarks

Results are cached. Call with `refreshCache` set to [true](#) if renderers change at runtime.

GetCapability<T>()

Casts the current avatar object to `T`.

```
public T GetCapability<T>() where T : ICapability
```

Returns

`T`

The casted result.

Type Parameters

`T`

The capability type to cast to.

ImporterSpecificDispose()

```
protected virtual void ImporterSpecificDispose()
```

RegisterLifetimeObject(Object)

Registers a Unity object to be destroyed when the avatar is disposed.

```
public void RegisterLifetimeObject(UnityEngine.Object unityObject)
```

Parameters

unityObject Object

Remarks

Intended for post-processors and extensions that create temporary runtime objects.

ThrowIfDisposed()

```
protected void ThrowIfDisposed()
```

TryGetAvatarMaterials()

Tries to get all materials of the avatar.

```
public virtual IReadOnlyList<Material>? TryGetAvatarMaterials()
```

Returns

IReadOnlyList<Material>

TryGetAvatarMeshes()

Tries to get all meshes of the avatar.

```
public virtual IReadOnlyList<Mesh>? TryGetAvatarMeshes()
```

Returns

IReadOnlyList<Mesh>

TryGetAvatarRenderers()

Tries to get all renderers of the avatar.

```
public virtual IReadOnlyList<Renderer>? TryGetAvatarRenderers()
```

Returns

`ReadOnlyList<Renderer>`

TryGetCapability<T>(out T?)

Tries to cast the current avatar object to `T`.

```
public bool TryGetCapability<T>(out T? capability) where T : ICapability
```

Parameters

`capability` `T`

The casted result, or [null](#) if casting failed.

Returns

`bool`

[true](#) if casted successfully; [false](#) otherwise.

Type Parameters

`T`

The capability type to cast to.

Remarks

Ultimately, this is just syntactic sugar for:

```
if (avatar is T capability)
    ...
else
    ...
```

Namespace Uralstech.AvLoader.Capabilities

Classes

[BlendShapeProviderExtensions](#)

Extensions for [IBlendShapeProviders](#).

Interfaces

[IAvatarExpressionProvider](#)

Provides semantically meaningful avatar expressions (e.g. Smile, Blink, Angry).

[IBlendShapeProvider](#)

Provides access to weighted blend channels.

[ICapability](#)

Base interface for all avatar capabilities.

[IImportedAnimator](#)

Exposes an [Animator](#) that is created by the avatar importer.

[IMeshBlendShapeProvider](#)

Provides access to mesh-level blendshapes.

Class BlendShapeProviderExtensions

Namespace: [Uralstech.AvLoader.Capabilities](#)

Extensions for [IBlendShapeProviders](#).

```
public static class BlendShapeProviderExtensions
```

Inheritance

object ← BlendShapeProviderExtensions

Methods

HasAnyWeight(IBlendShapeProvider, string[], out string?)

Checks if a channel with any one of the given **names** exists in the current provider.

```
public static bool HasAnyWeight(this IBlendShapeProvider current, string[] names, out string? foundName)
```

Parameters

current [IBlendShapeProvider](#)

names string[]

foundName string

The name of the found channel or [null](#) if not found.

Returns

bool

[true](#) if a channel was found; [false](#) otherwise.

Interface IAvatarExpressionProvider

Namespace: [Uralstech.AvLoader.Capabilities](#)

Provides semantically meaningful avatar expressions (e.g. Smile, Blink, Angry).

```
public interface IAvatarExpressionProvider : IBlendShapeProvider, ICapability
```

Inherited Members

[IBlendShapeProvider.ChannelNames](#) , [IBlendShapeProvider.HasWeights](#) ,
[IBlendShapeProvider.GetWeight\(string\)](#) , [IBlendShapeProvider.SetWeight\(string, float\)](#) ,
[IBlendShapeProvider.HasWeight\(string\)](#).

Extension Methods

[BlendShapeProviderExtensions.HasAnyWeight\(IBlendShapeProvider, string\[\], out string?\)](#).

Remarks

Expression names are avatar-level concepts and may drive multiple blendshapes, bones, or other mechanisms.

Interface IBlendShapeProvider

Namespace: [Uralstech.AvLoader.Capabilities](#)

Provides access to weighted blend channels.

```
public interface IBlendShapeProvider : ICapability
```

Extension Methods

[BlendShapeProviderExtensions.HasAnyWeight\(IBlendShapeProvider, string\[\], out string?\)](#)

Remarks

This interface does not imply any semantic meaning of the channels. Channel names and behavior are implementation-defined.

Properties

ChannelNames

Case-sensitive keys for all available blend weights.

```
ICollection<string> ChannelNames { get; }
```

Property Value

ICollection<string>

HasWeights

Does this avatar have any blendshape weights?

```
bool HasWeights { get; }
```

Property Value

bool

Methods

GetWeight(string)

Gets the weight for the given name.

```
float GetWeight(string name)
```

Parameters

name string

Returns

float

Remarks

If multiple blend shape sources in the avatar have the same channel name, this gets the maximum weight.

HasWeight(string)

Checks if a weight channel with the given name exists.

```
bool HasWeight(string name)
```

Parameters

name string

Returns

bool

SetWeight(string, float)

Sets the weight for the given name.

```
void SetWeight(string name, float weight)
```

Parameters

name string

weight float

Remarks

If multiple blend shape sources in the avatar have the same channel name, this sets the weight for all of them.

Interface ICapability

Namespace: [Uralstech.AvLoader.Capabilities](#)

Base interface for all avatar capabilities.

```
public interface ICapability
```

Interface IImportedAnimator

Namespace: [Uralstech.AvLoader.Capabilities](#)

Exposes an [Animator](#) that is created by the avatar importer.

```
public interface IImportedAnimator : ICapability
```

Remarks

This capability represents importer- or runtime-library-defined animation support. For example, UniVRM generates an `UnityEngine.Animator` as part of the import process.

Animators added later by post-processors or user code are not considered part of this capability and should be accessed directly from [GameObject](#).

Properties

Animator

The importer-generated animator.

```
Animator Animator { get; }
```

Property Value

Animator

Interface IMeshBlendShapeProvider

Namespace: [Uralstech.AvLoader.Capabilities](#)

Provides access to mesh-level blendshapes.

```
public interface IMeshBlendShapeProvider : IBlendShapeProvider, ICapability
```

Inherited Members

[IBlendShapeProvider.ChannelNames](#) , [IBlendShapeProvider.HasWeights](#) ,
[IBlendShapeProvider.GetWeight\(string\)](#) , [IBlendShapeProvider.SetWeight\(string, float\)](#) ,
[IBlendShapeProvider.HasWeight\(string\)](#).

Extension Methods

[BlendShapeProviderExtensions.HasAnyWeight\(IBlendShapeProvider, string\[\], out string?\)](#).

Remarks

Names correspond to mesh blendshape channels and are not guaranteed to represent avatar expressions.

Namespace Uralstech.AvLoader.DataLoaders

Classes

[FileAvDataLoader](#)

Loads avatars from on-device files.

[URIAvDataLoader](#)

Loads avatar data from remote URIs via `UnityWebRequest.Get(Uri)`. Downloads are performed in parallel.

Class FileAvDataLoader

Namespace: [Uralstech.AvLoader.DataLoaders](#)

Loads avatars from on-device files.

```
public class FileAvDataLoader : IAvDataLoader
```

Inheritance

object ← FileAvDataLoader

Implements

[IAvDataLoader](#)

Constructors

FileAvDataLoader(string, string, string?, string?)

Creates a loader configured to load avatar data from paths to each file.

```
public FileAvDataLoader(string modelFilePath, string metadataFilePath, string?  
fullRenderFilePath = null, string? bustRenderFilePath = null)
```

Parameters

modelFilePath string

Path to the model.

metadataFilePath string

Path to the avatar metadata.

fullRenderFilePath string

Path to the full render of the avatar (optional).

bustRenderFilePath string

Path to the bust render of the avatar (optional).

Exceptions

NotSupportedException

Thrown if the file extension of `modelFilePath`, `fullRenderFilePath` or `bustRenderFilePath` is not supported.

FileAvDataLoader(string, AvModelFileExtension, AvImageFileExtension, AvImageFileExtension, bool, bool)

Creates a loader configured to load avatar data from the directory at `basePath`.

```
public FileAvDataLoader(string basePath, AvModelFileExtension modelFormat =  
    AvModelFileExtension.None, AvImageFileExtension fullRenderFormat =  
    AvImageFileExtension.None, AvImageFileExtension bustRenderFormat =  
    AvImageFileExtension.None, bool shouldLoadFullRender = false, bool shouldLoadBustRender  
    = false)
```

Parameters

`basePath` string

The base directory containing the avatar and its metadata.

`modelFormat` [AvModelFileExtension](#)

The expected file extension(s) of the model, leave as [None](#) to auto detect of all supported formats.

`fullRenderFormat` [AvImageFileExtension](#)

The expected file extension(s) of the model's full render, leave as [None](#) to auto detect of all supported formats.

`bustRenderFormat` [AvImageFileExtension](#)

The expected file extension(s) of the model's bust render, leave as [None](#) to auto detect of all supported formats.

`shouldLoadFullRender` bool

Should the full render be loaded?

`shouldLoadBustRender` bool

Should the bust render be loaded?

Remarks

basePath **must** contain the avatar and associated data in the following structure:

```
basePath/  
├─ model.[supported model extension]      # The avatar's model  
├─ metadata.json                          # Metadata  
├─ full.[supported image extension]        # Full render of the avatar (optional)  
└─ bust.[supported image extension]        # Bust render of the avatar (optional)
```

If you want to use a different file structure, call [FileAvDataLoader\(string, string, string?, string?\)](#) instead.

Methods

LoadAvatarAsync(bool, CancellationToken)

Tries loading an avatar into an [AvSourceData](#).

```
public Awaitable<AvSourceData?> LoadAvatarAsync(bool throwOnFail, CancellationToken token  
= default)
```

Parameters

throwOnFail bool

Should this method throw errors on failures or log them as warnings and return [null](#)?

token CancellationToken

Returns

Awaitable<[AvSourceData](#)>

The loaded data if successful; [null](#) on failure.

TrySearchForBustRender(string, AvImageFileExtension, out string?, out AvImageFileExtension)

Searches for a supported bust-render image file inside **directory** using the default bust-render file name.

```
public static bool TrySearchForBustRender(string directory, AvImageFileExtension
searchFilter, out string? foundPath, out AvImageFileExtension foundExtension)
```

Parameters

directory string

The directory to search in.

searchFilter [AvImageFileExtension](#)

The expected image file extension(s). Use [None](#) or grouped values to search across all supported formats.

foundPath string

When this method returns [true](#), contains the full path to the found image file.

foundExtension [AvImageFileExtension](#)

When this method returns [true](#), contains the detected image file extension.

Returns

bool

[true](#) if a valid image file was found; otherwise, [false](#).

TrySearchForFullRender(string, AvImageFileExtension, out string?, out AvImageFileExtension)

Searches for a supported full-render image file inside **directory** using the default full-render file name.

```
public static bool TrySearchForFullRender(string directory, AvImageFileExtension
searchFilter, out string? foundPath, out AvImageFileExtension foundExtension)
```

Parameters

directory string

The directory to search in.

searchFilter [AvImageFileExtension](#)

The expected image file extension(s). Use [None](#) or grouped values to search across all supported formats.

foundPath string

When this method returns [true](#), contains the full path to the found image file.

foundExtension [AvImageFileExtension](#)

When this method returns [true](#), contains the detected image file extension.

Returns

bool

[true](#) if a valid image file was found; otherwise, [false](#).

TrySearchForImageFile(string, string, AvImageFileExtension, out string?, out AvImageFileExtension)

Searches for a supported image file inside **directory** using a custom base file name.

```
public static bool TrySearchForImageFile(string directory, string
fileNameWithoutExtensino, AvImageFileExtension searchFilter, out string? foundPath, out
AvImageFileExtension foundExtension)
```

Parameters

directory string

The directory to search in.

fileNameWithoutExtensino string

The image file name without an extension.

searchFilter [AvImageFileExtension](#)

The expected image file extension(s). Use [None](#) or grouped values to search across all supported formats.

foundPath string

When this method returns [true](#), contains the full path to the found image file.

foundExtension [AvImageFileExtension](#)

When this method returns [true](#), contains the detected image file extension.

Returns

bool

[true](#) if a valid image file was found; otherwise, [false](#).

TrySearchForModelFile(string, string, AvModelFileExtension, out string?, out AvModelFileExtension)

Searches for a supported avatar model file inside **directory** using a custom base file name.

```
public static bool TrySearchForModelFile(string directory, string
fileNameWithoutExtensino, AvModelFileExtension searchFilter, out string? foundPath, out
AvModelFileExtension foundExtension)
```

Parameters

directory string

The directory to search in.

fileNameWithoutExtensino string

The model file name without an extension.

searchFilter [AvModelFileExtension](#)

The expected model file extension(s). Use [None](#) or grouped values to search across all supported formats.

foundPath string

When this method returns [true](#), contains the full path to the found model file.

foundExtension [AvModelFileExtension](#)

When this method returns [true](#), contains the detected model file extension.

Returns

bool

[true](#) if a valid model file was found; otherwise, [false](#).

TrySearchForModelFile(string, AvModelFileExtension, out string?, out AvModelFileExtension)

Searches for a supported avatar model file inside **directory** using the default model file name.

```
public static bool TrySearchForModelFile(string directory, AvModelFileExtension
searchFilter, out string? foundPath, out AvModelFileExtension foundExtension)
```

Parameters

directory string

The directory to search in.

searchFilter [AvModelFileExtension](#)

The expected model file extension(s). Use [None](#) or grouped values to search across all supported formats.

foundPath string

When this method returns [true](#), contains the full path to the found model file.

foundExtension [AvModelFileExtension](#)

When this method returns [true](#), contains the detected model file extension.

Returns

bool

[true](#) if a valid model file was found; otherwise, [false](#).

Class URIAvDataLoader

Namespace: [Uralstech.AvLoader.DataLoaders](#)

Loads avatar data from remote URIs via UnityWebRequest.Get(Uri). Downloads are performed in parallel.

```
public class URIAvDataLoader : IAvDataLoader
```

Inheritance

object ← URIAvDataLoader

Implements

[IAvDataLoader](#)

Constructors

URIAvDataLoader(string, string, AvModelFileExtension, string?, string?, Encoding?)

Creates a loader configured to download avatar data from explicit URIs.

```
public URIAvDataLoader(string modelURI, string metadataURI, AvModelFileExtension  
modelFormat, string? fullRenderURI = null, string? bustRenderURI = null, Encoding?  
metadataEncoding = null)
```

Parameters

modelURI string

Direct URI to the avatar model file.

metadataURI string

Direct URI to the metadata JSON file.

modelFormat [AvModelFileExtension](#)

The file format/extension of the model. Must be explicitly specified (no auto-detection like with [File AvDataLoader](#)).

fullRenderURI string

Optional URI to the full-body render image.

bustRenderURI string

Optional URI to the bust/half-body render image.

metadataEncoding Encoding

Text encoding for the metadata JSON. Defaults to UTF-8.

Exceptions

ArgumentException

Thrown if **modelFormat** is [None](#), as URI-based loading requires an explicit format.

URIAvDataLoader(Uri, Uri, AvModelFileExtension, Uri?, Uri?, Encoding?)

Creates a loader configured to download avatar data from explicit URIs.

```
public URIAvDataLoader(Uri modelURI, Uri metadataURI, AvModelFileExtension modelFormat, Uri? fullRenderURI = null, Uri? bustRenderURI = null, Encoding? metadataEncoding = null)
```

Parameters

modelURI Uri

Direct URI to the avatar model file.

metadataURI Uri

Direct URI to the metadata JSON file.

modelFormat [AvModelFileExtension](#)

The file format/extension of the model. Must be explicitly specified (no auto-detection like with [File AvDataLoader](#)).

fullRenderURI Uri

Optional URI to the full-body render image.

bustRenderURI Uri

Optional URI to the bust/half-body render image.

metadataEncoding Encoding

Text encoding for the metadata JSON. Defaults to UTF-8.

Exceptions

ArgumentException

Thrown if **modelFormat** is [None](#), as URI-based loading requires an explicit format.

Fields

WebRequestConfigurationCallback

Called each time a new UnityWebRequest is created, e.g. for auth configuration.

```
public Func<UnityWebRequest, Awaitable>? WebRequestConfigurationCallback
```

Field Value

Func<UnityWebRequest, Awaitable>

Methods

LoadAvatarAsync(bool, CancellationToken)

Tries loading an avatar into an [AvSourceData](#).

```
public Awaitable<AvSourceData?> LoadAvatarAsync(bool throwOnFail, CancellationToken token  
= default)
```

Parameters

`throwOnFail` bool

Should this method throw errors on failures or log them as warnings and return [null](#)?

`token` CancellationToken

Returns

Awaitable<[AvSourceData](#)>

The loaded data if successful; [null](#) on failure.

Namespace Uralstech.AvLoader.Importers

Classes

[GLTFastAvImporter](#)

Imports glTF avatars using [glTFast](#).

[LoadedGLTFastAv](#)

A loaded glTFast glTF avatar.

[LoadedUniGLTFAv](#)

A loaded UniGLTF glTF avatar.

[LoadedUniVRM10Av](#)

A loaded UniVRM VRM10 avatar.

[UniGLTFAvImporter](#)

Imports glTF avatars using [UniGLTF](#).

[UniVRM10AvImporter](#)

Imports VRM avatars using [UniVRM10](#).

Class GLTFastAvImporter

Namespace: [Uralstech.AvLoader.Importers](#)

Imports glTF avatars using [glTFast](#).

```
public class GLTFastAvImporter : IAvImporter
```

Inheritance

object ← GLTFastAvImporter

Implements

[IAvImporter](#)

Fields

DeferAgent

Optional custom glTFast defer agent.

```
public IDeferAgent? DeferAgent
```

Field Value

IDEferAgent?

DownloadProvider

Optional custom glTFast download provider.

```
public IDownloadProvider? DownloadProvider
```

Field Value

IDownloadProvider?

ImportSettings

Optional custom glTFast ImportSettings.

```
public ImportSettings? ImportSettings
```

Field Value

ImportSettings?

InstantiationSettings

Optional custom glTFast InstantiationSettings.

```
public InstantiationSettings? InstantiationSettings
```

Field Value

InstantiationSettings?

InstantiatorFactory

Optional factory for custom glTFast GameObjectInstantiators.

```
public Func<GltfImport, GameObject, Awaitable<GameObjectInstantiator>>? InstantiatorFactory
```

Field Value

Func<GltfImport, GameObject, Awaitable<GameObjectInstantiator>>

Logger

Optional custom glTFast logger.

```
public ICodeLogger? Logger
```

Field Value

ILogger?

MaterialGenerator

Optional custom glTF material generator.

```
public ILogger? MaterialGenerator
```

Field Value

MaterialGenerator?

Methods

ImportAvatarAsync(AvSourceData, bool, CancellationToken)

Tries to import the avatar into a scene as a disabled GameObject, along with metadata and any renders.

```
public Awaitable<LoadedAv?> ImportAvatarAsync(AvSourceData rawData, bool throwOnFail,
    CancellationToken token = default)
```

Parameters

rawData [AvSourceData](#)

The raw avatar data to process.

throwOnFail bool

Should this method throw errors on failures or log them as warnings and return [null](#)?

token CancellationToken

Returns

Awaitable<[LoadedAv](#)>

The loaded avatar if successful; [null](#) on failure.

Remarks

The returned [LoadedAv](#) may contain code to handle requirements of format-supporting plugins. For example, [GLTFastAvImporter](#) depends on glTFast, which requires that GltfImport object used to import the avatar be active alongside the avatar's GameObject, and should be disposed after the avatar is no longer needed. Thus, [LoadedAv](#) implements System.IDisposable.

SupportsFormat(AvModelFileExtension)

Returns [true](#) if this importer can handle the given format; [false](#) otherwise.

```
public bool SupportsFormat(AvModelFileExtension format)
```

Parameters

format [AvModelFileExtension](#)

Returns

bool

Class LoadedGLTFastAv

Namespace: [Uralstech.AvLoader.Importers](#)

A loaded glTF glTF avatar.

```
public class LoadedGLTFastAv : LoadedAv, IMeshBlendShapeProvider,
    IBlendShapeProvider, ICapability
```

Inheritance

object ← [LoadedAv](#) ← LoadedGLTFastAv

Implements

[IMeshBlendShapeProvider](#), [IBlendShapeProvider](#), [ICapability](#)

Inherited Members

[LoadedAv.GameObject](#), [LoadedAv.Metadata](#), [LoadedAv.FullRender](#), [LoadedAv.BustRender](#),
[LoadedAv.DestroyOnDispose](#), [LoadedAv.LifetimeObjects](#), [LoadedAv.ImporterType](#),
[LoadedAv.RegisterLifetimeObject\(UnityEngine.Object\)](#),
[LoadedAv.DeregisterLifetimeObject\(UnityEngine.Object\)](#), [LoadedAv.TryGetCapability<T>\(out T\)](#),
[LoadedAv.GetCapability<T>\(\)](#), [LoadedAv.GetAvatarMaterials\(bool\)](#),
[LoadedAv.GetAvatarRenderers\(bool\)](#), [LoadedAv.GetAvatarMeshes\(bool\)](#),
[LoadedAv.TryGetAvatarRenderers\(\)](#), [LoadedAv.ThrowIfDisposed\(\)](#), [LoadedAv.Dispose\(\)](#)

Extension Methods

[BlendShapeProviderExtensions.HasAnyWeight\(IBlendShapeProvider, string\[\], out string?\)](#)

Constructors

LoadedGLTFastAv(GameObject, GltfImport, AvMetadata, Texture2D?, Texture2D?, Type)

```
public LoadedGLTFastAv(GameObject gameObject, GltfImport import, AvMetadata metadata,
    Texture2D? fullRender, Texture2D? bustRender, Type importerType)
```

Parameters

gameObject GameObject

import GltfImport

metadata [AvMetadata](#)

fullRender Texture2D?

bustRender Texture2D?

importerType Type

Fields

Import

The glTFast import associated with the avatar.

```
public readonly GltfImport Import
```

Field Value

GltfImport

Properties

ChannelNames

Case-sensitive keys for all available blend weights.

```
public IReadOnlyCollection<string> ChannelNames { get; }
```

Property Value

IReadOnlyCollection<string>

HasWeights

Does this avatar have any blendshape weights?

```
public bool HasWeights { get; }
```

Property Value

bool

Methods

GetWeight(string)

Gets the weight for the given name.

```
public float GetWeight(string name)
```

Parameters

name string

Returns

float

Remarks

If multiple blend shape sources in the avatar have the same channel name, this gets the maximum weight.

HasWeight(string)

Checks if a weight channel with the given name exists.

```
public bool HasWeight(string name)
```

Parameters

name string

Returns

bool

ImporterSpecificDispose()

```
protected override void ImporterSpecificDispose()
```

SetWeight(string, float)

Sets the weight for the given name.

```
public void SetWeight(string name, float weight)
```

Parameters

name string

weight float

Remarks

If multiple blend shape sources in the avatar have the same channel name, this sets the weight for all of them.

TryGetAvatarMaterials()

Tries to get all materials of the avatar.

```
public override IReadOnlyList<Material>? TryGetAvatarMaterials()
```

Returns

IReadOnlyList<Material>

TryGetAvatarMeshes()

Tries to get all meshes of the avatar.

```
public override IReadOnlyList<Mesh>? TryGetAvatarMeshes()
```

Returns

IReadOnlyList<Mesh>

Class LoadedUniGLTFAv

Namespace: [Uralstech.AvLoader.Importers](#)

A loaded UniGLTF glTF avatar.

```
public class LoadedUniGLTFAv : LoadedAv, IMeshBlendShapeProvider,
    IBlendShapeProvider, ICapability
```

Inheritance

object ← [LoadedAv](#) ← LoadedUniGLTFAv

Implements

[IMeshBlendShapeProvider](#), [IBlendShapeProvider](#), [ICapability](#)

Inherited Members

[LoadedAv.GameObject](#), [LoadedAv.Metadata](#), [LoadedAv.FullRender](#), [LoadedAv.BustRender](#),
[LoadedAv.DestroyOnDispose](#), [LoadedAv.LifetimeObjects](#), [LoadedAv.ImporterType](#),
[LoadedAv.RegisterLifetimeObject\(UnityEngine.Object\)](#),
[LoadedAv.DeregisterLifetimeObject\(UnityEngine.Object\)](#), [LoadedAv.TryGetCapability<T>\(out T\)](#),
[LoadedAv.GetCapability<T>\(\)](#), [LoadedAv.GetAvatarMaterials\(bool\)](#),
[LoadedAv.GetAvatarRenderers\(bool\)](#), [LoadedAv.GetAvatarMeshes\(bool\)](#), [LoadedAv.ThrowIfDisposed\(\)](#),
[LoadedAv.Dispose\(\)](#).

Extension Methods

[BlendShapeProviderExtensions.HasAnyWeight\(IBlendShapeProvider, string\[\], out string?\)](#)

Constructors

LoadedUniGLTFAv(GameObject, RuntimeGltfInstance, AvMetadata, Texture2D?, Texture2D?, Type)

```
public LoadedUniGLTFAv(GameObject gameObject, RuntimeGltfInstance gltfInstance, AvMetadata
    metadata, Texture2D? fullRender, Texture2D? bustRender, Type importerType)
```

Parameters

gameObject GameObject

gltfInstance RuntimeGltfInstance

metadata [AvMetadata](#)

fullRender Texture2D?

bustRender Texture2D?

importerType Type

Fields

GLTFInstance

The glTF avatar.

```
public readonly RuntimeGltfInstance GLTFInstance
```

Field Value

RuntimeGltfInstance

Properties

ChannelNames

Case-sensitive keys for all available blend weights.

```
public IReadOnlyCollection<string> ChannelNames { get; }
```

Property Value

IReadOnlyCollection<string>

HasWeights

Does this avatar have any blendshape weights?

```
public bool HasWeights { get; }
```

Property Value

bool

Methods

GetWeight(string)

Gets the weight for the given name.

```
public float GetWeight(string name)
```

Parameters

name string

Returns

float

Remarks

If multiple blend shape sources in the avatar have the same channel name, this gets the maximum weight.

HasWeight(string)

Checks if a weight channel with the given name exists.

```
public bool HasWeight(string name)
```

Parameters

name string

Returns

bool

ImporterSpecificDispose()

```
protected override void ImporterSpecificDispose()
```

SetWeight(string, float)

Sets the weight for the given name.

```
public void SetWeight(string name, float weight)
```

Parameters

name string

weight float

Remarks

If multiple blend shape sources in the avatar have the same channel name, this sets the weight for all of them.

TryGetAvatarMaterials()

Tries to get all materials of the avatar.

```
public override IReadOnlyList<Material>? TryGetAvatarMaterials()
```

Returns

IReadOnlyList<Material>

TryGetAvatarMeshes()

Tries to get all meshes of the avatar.

```
public override IReadOnlyList<Mesh>? TryGetAvatarMeshes()
```

Returns

IReadOnlyList<Mesh>

TryGetAvatarRenderers()

Tries to get all renderers of the avatar.

```
public override IReadOnlyList<Renderer>? TryGetAvatarRenderers()
```

Returns

IReadOnlyList<Renderer>

Class LoadedUniVRM10Av

Namespace: [Uralstech.AvLoader.Importers](#)

A loaded UniVRM VRM10 avatar.

```
public class LoadedUniVRM10Av : LoadedAv, IImportedAnimator, IAvatarExpressionProvider,
    IBlendShapeProvider, ICapability
```

Inheritance

object ← [LoadedAv](#) ← LoadedUniVRM10Av

Implements

[IImportedAnimator](#), [IAvatarExpressionProvider](#), [IBlendShapeProvider](#), [ICapability](#)

Inherited Members

[LoadedAv.GameObject](#), [LoadedAv.Metadata](#), [LoadedAv.FullRender](#), [LoadedAv.BustRender](#),
[LoadedAv.DestroyOnDispose](#), [LoadedAv.LifetimeObjects](#), [LoadedAv.ImporterType](#),
[LoadedAv.RegisterLifetimeObject\(UnityEngine.Object\)](#),
[LoadedAv.DeregisterLifetimeObject\(UnityEngine.Object\)](#), [LoadedAv.TryGetCapability<T>\(out T\)](#),
[LoadedAv.GetCapability<T>\(\)](#), [LoadedAv.GetAvatarMaterials\(bool\)](#),
[LoadedAv.GetAvatarRenderers\(bool\)](#), [LoadedAv.GetAvatarMeshes\(bool\)](#), [LoadedAv.ThrowIfDisposed\(\)](#),
[LoadedAv.Dispose\(\)](#).

Extension Methods

[BlendShapeProviderExtensions.HasAnyWeight\(IBlendShapeProvider, string\[\], out string?\)](#)

Constructors

LoadedUniVRM10Av(GameObject, Vrm10Instance, AvMetadata, Texture2D?, Texture2D?, Type)

```
public LoadedUniVRM10Av(GameObject gameObject, Vrm10Instance vrmInstance, AvMetadata
    metadata, Texture2D? fullRender, Texture2D? bustRender, Type importerType)
```

Parameters

gameObject GameObject

vrmlInstance Vrm10Instance

metadata [AvMetadata](#)

fullRender Texture2D?

bustRender Texture2D?

importerType Type

Fields

GLTFInstance

The RuntimeGltfInstance component of the VRM avatar, if it exists.

```
public readonly RuntimeGltfInstance? GLTFInstance
```

Field Value

RuntimeGltfInstance?

VRMInstance

The VRM avatar.

```
public readonly Vrm10Instance VRMInstance
```

Field Value

Vrm10Instance

Properties

Animator

The importer-generated animator.

```
public Animator Animator { get; }
```

Property Value

Animator

ChannelNames

Case-sensitive keys for all available blend weights.

```
public IReadOnlyCollection<string> ChannelNames { get; }
```

Property Value

IReadOnlyCollection<string>

HasWeights

Does this avatar have any blendshape weights?

```
public bool HasWeights { get; }
```

Property Value

bool

Methods

GetWeight(string)

Gets the weight for the given name.

```
public float GetWeight(string name)
```

Parameters

name string

Returns

float

Remarks

If multiple blend shape sources in the avatar have the same channel name, this gets the maximum weight.

HasWeight(string)

Checks if a weight channel with the given name exists.

```
public bool HasWeight(string name)
```

Parameters

name string

Returns

bool

ImporterSpecificDispose()

```
protected override void ImporterSpecificDispose()
```

SetWeight(string, float)

Sets the weight for the given name.

```
public void SetWeight(string name, float weight)
```

Parameters

name string

weight float

Remarks

If multiple blend shape sources in the avatar have the same channel name, this sets the weight for all of them.

TryGetAvatarMaterials()

Tries to get all materials of the avatar.

```
public override IReadOnlyList<Material>? TryGetAvatarMaterials()
```

Returns

IReadOnlyList<Material>

TryGetAvatarMeshes()

Tries to get all meshes of the avatar.

```
public override IReadOnlyList<Mesh>? TryGetAvatarMeshes()
```

Returns

IReadOnlyList<Mesh>

TryGetAvatarRenderers()

Tries to get all renderers of the avatar.

```
public override IReadOnlyList<Renderer>? TryGetAvatarRenderers()
```

Returns

IReadOnlyList<Renderer>

Class UniGLTFAvImporter

Namespace: [Uralstech.AvLoader.Importers](#)

Imports glTF avatars using [UniGLTF](#).

```
public class UniGLTFAvImporter : IAvImporter
```

Inheritance

object ← UniGLTFAvImporter

Implements

[IAvImporter](#)

Fields

EnableUpdateWhenOffscreen

Should the avatar's SkinnedMeshRenderers be updated when off-screen?

```
public bool EnableUpdateWhenOffscreen
```

Field Value

bool

ImporterContextSettings

Optional importer settings.

```
public ImporterContextSettings? ImporterContextSettings
```

Field Value

ImporterContextSettings?

MaterialGenerator

Optional material generator.

```
public IMaterialDescriptorGenerator? MaterialGenerator
```

Field Value

IMaterialDescriptorGenerator?

ShowMeshes

Should all mesh renderers be enabled on load? (default: [true](#) )

```
public bool ShowMeshes
```

Field Value

bool

TextureDeserializer

Optional texture deserialization.

```
public ITextureDeserializer? TextureDeserializer
```

Field Value

ITextureDeserializer?

Methods

ImportAvatarAsync(AvSourceData, bool, CancellationToken)

Tries to import the avatar into a scene as a disabled GameObject, along with metadata and any renders.

```
public Awaitable<LoadedAv?> ImportAvatarAsync(AvSourceData rawData, bool throwOnFail,
CancellationTokentoken = default)
```

Parameters

rawData [AvSourceData](#)

The raw avatar data to process.

throwOnFail bool

Should this method throw errors on failures or log them as warnings and return [null](#)?

token CancellationToken

Returns

Awaitable<[LoadedAv](#)>

The loaded avatar if successful; [null](#) on failure.

Remarks

The returned [LoadedAv](#) may contain code to handle requirements of format-supporting plugins. For example, [GLTFastAvImporter](#) depends on glTFast, which requires that GltfImport object used to import the avatar be active alongside the avatar's GameObject, and should be disposed after the avatar is no longer needed. Thus, [LoadedAv](#) implements System.IDisposable.

SupportsFormat(AvModelFileExtension)

Returns [true](#) if this importer can handle the given format; [false](#) otherwise.

```
public bool SupportsFormat(AvModelFileExtension format)
```

Parameters

format [AvModelFileExtension](#)

Returns

bool

Class UniVRM10AvImporter

Namespace: [Uralstech.AvLoader.Importers](#)

Imports VRM avatars using [UniVRM10](#).

```
public class UniVRM10AvImporter : IAvImporter
```

Inheritance

object ← UniVRM10AvImporter

Implements

[IAvImporter](#)

Fields

CanLoadVrm0X

If [true](#), VRM-0.x models are converted to VRM-1.0 and loaded. The components attached to the VRM-0.x model are different. (default: [true](#))

```
public bool CanLoadVrm0X
```

Field Value

bool

ControlRigGenerationOption

ControlRig generation settings. (default: ControlRigGenerationOption.Generate)

```
public ControlRigGenerationOption ControlRigGenerationOption
```

Field Value

ControlRigGenerationOption

ImporterContextSettings

Optional importer settings.

```
public ImporterContextSettings? ImporterContextSettings
```

Field Value

ImporterContextSettings?

MaterialGenerator

Optional material generator.

```
public IMaterialDescriptorGenerator? MaterialGenerator
```

Field Value

IMaterialDescriptorGenerator?

ShowMeshes

Should all mesh renderers be enabled on load? (default: [true](#) )

```
public bool ShowMeshes
```

Field Value

bool

SpringboneRuntime

Optional runtime selection for SpringBones.

```
public IVrm10SpringBoneRuntime? SpringboneRuntime
```

Field Value

IVrm10SpringBoneRuntime?

TextureDeserializer

Optional texture deserialization.

```
public ITextureDeserializer? TextureDeserializer
```

Field Value

ITextureDeserializer?

UpdateType

Update type of the loaded avatar.

```
public Vrm10Instance.UpdateTypes UpdateType
```

Field Value

UpdateTypes

VRMMetaInformationCallback

Optional callback called on VRM-0x to VRM-1.0 migration.

```
public Vrm10.VrmMetaInformationCallback? VRMMetaInformationCallback
```

Field Value

Vrm10.VrmMetaInformationCallback?

Methods

ImportAvatarAsync(AvSourceData, bool, CancellationToken)

Tries to import the avatar into a scene as a disabled GameObject, along with metadata and any renders.

```
public Awaitable<LoadedAv?> ImportAvatarAsync(AvSourceData rawData, bool throwOnFail,
CancellationToken token = default)
```

Parameters

rawData [AvSourceData](#)

The raw avatar data to process.

throwOnFail bool

Should this method throw errors on failures or log them as warnings and return [null](#)?

token CancellationToken

Returns

Awaitable<[LoadedAv](#)>

The loaded avatar if successful; [null](#) on failure.

Remarks

The returned [LoadedAv](#) may contain code to handle requirements of format-supporting plugins. For example, [GLTFastAvImporter](#) depends on glTFast, which requires that GltfImport object used to import the avatar be active alongside the avatar's GameObject, and should be disposed after the avatar is no longer needed. Thus, [LoadedAv](#) implements System.IDisposable.

SupportsFormat(AvModelFileExtension)

Returns [true](#) if this importer can handle the given format; [false](#) otherwise.

```
public bool SupportsFormat(AvModelFileExtension format)
```

Parameters

format [AvModelFileExtension](#)

Returns

bool

Namespace Uralstech.AvLoader.PostProcessors

Classes

[ActionPostProcessor](#)

QoL post-processor that runs an [Action](#).

[AnimatorPostProcessor](#)

Post-processing step which adds or updates an animation controller in the loaded avatar. Requires Unity's animation module to be enabled.

[AsyncFuncPostProcessor](#)

QoL post-processor that runs an async [Func](#).

[AvPost](#)

Fluent helper methods for creating common avatar post-processors.

[CachePostProcessor](#)

Post-processing step that saves the loaded avatar's data to a local directory for future use. Skips execution if the avatar was originally loaded using [FileAvDataLoader](#), unless overridden via [IgnoreOriginalLoaderType](#).

[ConditionalPostProcessor](#)

Post-processing step that runs based on the given condition.

[MaterialOverrideConfig](#)

Configuration data for [MaterialOverridePostProcessor](#).

[MaterialOverridePostProcessor](#)

Post-processing step that recreates avatar materials using alternative shaders and copies selected property and keyword values from the original materials.

[ShaderSwapConfig](#)

Configuration data for [ShaderSwapPostProcessor](#).

[ShaderSwapPostProcessor](#)

Post-processing step that swaps the avatar's shaders with the provided set of shaders.

Interfaces

[IAvatarDependentComponent](#)

Marks a Component as semantically dependent on a [LoadedAv](#) instance.

Class ActionPostProcessor

Namespace: [Uralstech.AvLoader.PostProcessors](#)

QoL post-processor that runs an [Action](#).

```
public class ActionPostProcessor : IAvPostProcessor
```

Inheritance

object ← ActionPostProcessor

Implements

[IAvPostProcessor](#)

Constructors

ActionPostProcessor(Action<LoadedAv, AvSourceData>)

```
public ActionPostProcessor(Action<LoadedAv, AvSourceData> action)
```

Parameters

action Action<[LoadedAv](#), [AvSourceData](#)>

Fields

Action

The action to execute.

```
public Action<LoadedAv, AvSourceData> Action
```

Field Value

Action<[LoadedAv](#), [AvSourceData](#)>

Methods

PostProcess(LoadedAv, AvSourceData)

Runs a post-processing step on the given avatar.

```
public void PostProcess(LoadedAv avatar, AvSourceData rawData)
```

Parameters

avatar [LoadedAv](#)

The loaded avatar to process.

rawData [AvSourceData](#)

The raw data of the loaded avatar.

Class AnimatorPostProcessor

Namespace: [Uralstech.AvLoader.PostProcessors](#)

Post-processing step which adds or updates an animation controller in the loaded avatar. Requires Unity's animation module to be enabled.

```
public class AnimatorPostProcessor : IAvPostProcessor
```

Inheritance

object ← AnimatorPostProcessor

Implements

[IAvPostProcessor](#)

Fields

AnimatorController

Runtime animator controller to assign to the loaded avatar.

```
public RuntimeAnimatorController? AnimatorController
```

Field Value

RuntimeAnimatorController?

AvGenderToAvatarLookup

Lookup table for when you need to assign animation avatars based on the avatar's gender. If not found here, falls back to [Avatar](#).

```
public IReadOnlyDictionary<AvGender, Avatar>? AvGenderToAvatarLookup
```

Field Value

IReadOnlyDictionary<[AvGender](#), Avatar>

Remarks

If the avatar already has an animator, this won't be assigned unless [OverrideAvatar](#) is [true](#)[↗].

Avatar

Animation avatar to assign to the loaded avatar.

```
public Avatar? Avatar
```

Field Value

Avatar?

Remarks

If the avatar already has an animator, this won't be assigned unless [OverrideAvatar](#) is [true](#)[↗].

OverrideAvatar

If the avatar already has an animation avatar assigned, should it be overridden?

```
public bool OverrideAvatar
```

Field Value

bool

Methods

PostProcess(LoadedAv, AvSourceData)

Runs a post-processing step on the given avatar.

```
public void PostProcess(LoadedAv avatar, AvSourceData _)
```

Parameters

avatar [LoadedAv](#)

The loaded avatar to process.

[_ AvSourceData](#)

Class AsyncFuncPostProcessor

Namespace: [Uralstech.AvLoader.PostProcessors](#)

QoL post-processor that runs an async [Func](#).

```
public class AsyncFuncPostProcessor : IAsyncAvPostProcessor
```

Inheritance

object ← AsyncFuncPostProcessor

Implements

[IAsyncAvPostProcessor](#)

Constructors

AsyncFuncPostProcessor(Func<LoadedAv, AvSourceData, CancellationTokens, Awaitable>)

```
public AsyncFuncPostProcessor(Func<LoadedAv, AvSourceData, CancellationTokens, Awaitable> func)
```

Parameters

func Func<[LoadedAv](#), [AvSourceData](#), CancellationTokens, Awaitable>

Fields

Func

The System.Func<T1, T2, T3, TResult> to execute.

```
public Func<LoadedAv, AvSourceData, CancellationTokens, Awaitable> Func
```

Field Value

Func<[LoadedAv](#), [AvSourceData](#), CancellationToken, Awaitable>

Methods

PostProcessAsync(LoadedAv, AvSourceData, CancellationToken)

Runs a post-processing step on the given avatar.

```
public Awaitable PostProcessAsync(LoadedAv avatar, AvSourceData rawData, CancellationToken token = default)
```

Parameters

avatar [LoadedAv](#)

The loaded avatar to process.

rawData [AvSourceData](#)

The raw data of the loaded avatar.

token CancellationToken

Returns

Awaitable

Class AvPost

Namespace: [Uralstech.AvLoader.PostProcessors](#)

Fluent helper methods for creating common avatar post-processors.

```
public static class AvPost
```

Inheritance

object ← AvPost

Methods

CacheTo(string, bool, AvImageFileExtension, AvImageFileExtension, int, int, bool)

Creates a post-processor that caches the loaded avatar's source data to disk.

```
public static CachePostProcessor CacheTo(string directory, bool useAvatarIdAsChildDirName = true, AvImageFileExtension fullRenderImageFormat = AvImageFileExtension.JPG, AvImageFileExtension bustRenderImageFormat = AvImageFileExtension.JPG, int fullRenderJPEGQuality = 100, int bustRenderJPEGQuality = 100, bool ignoreOriginalLoaderType = false)
```

Parameters

directory string

The base directory to save the cached avatar data to.

useAvatarIdAsChildDirName bool

If [true](#), the avatar is stored in a child directory named after its ID.

fullRenderImageFormat [AvImageFileExtension](#)

The image format to use when saving the full-body render.

bustRenderImageFormat [AvImageFileExtension](#)

The image format to use when saving the bust render.

fullRenderJPEGQuality int

JPEG quality to use when saving the full-body render.

bustRenderJPEGQuality int

JPEG quality to use when saving the bust render.

ignoreOriginalLoaderType bool

If [true](#), caching will occur even if the avatar was originally loaded from disk.

Returns

[CachePostProcessor](#)

Remarks

This only works for formats where the model does not depend on resources stored separately from the source file ([Model](#)). For example, avatars loaded from OBJ files with external texture files will not work.

ConfigureAnimator(RuntimeAnimatorController?, Avatar?, IReadOnlyDictionary<AvGender, Avatar>?, bool)

Creates a post-processor that configures an Animator on the loaded avatar.

```
public static AnimatorPostProcessor ConfigureAnimator(RuntimeAnimatorController?
animatorController = null, Avatar? avatar = null, IReadOnlyDictionary<AvGender, Avatar>?
avGenderToAvatarLookup = null, bool overrideAvatar = false)
```

Parameters

animatorController RuntimeAnimatorController?

The runtime animator controller to assign to the avatar.

avatar Avatar?

The animation avatar to assign if no gender-specific match is found.

avGenderToAvatarLookup IReadOnlyDictionary<[AvGender](#), Avatar>

Optional lookup table mapping avatar gender to animation avatars.

overrideAvatar bool

If [true](#), any existing animation avatar will be overridden.

Returns

[AnimatorPostProcessor](#)

Do(Action<LoadedAv, AvSourceData>)

Creates a post-processor that executes the given action.

```
public static ActionPostProcessor Do(Action<LoadedAv, AvSourceData> action)
```

Parameters

action Action<[LoadedAv](#), [AvSourceData](#)>

The action to execute during post-processing.

Returns

[ActionPostProcessor](#)

DoAsync(Func<LoadedAv, AvSourceData, CancellationToken, Awaitable>)

Creates an asynchronous post-processor that executes the given function.

```
public static AsyncFuncPostProcessor DoAsync(Func<LoadedAv, AvSourceData, CancellationToken, Awaitable> func)
```

Parameters

func Func<[LoadedAv](#), [AvSourceData](#), CancellationToken, Awaitable>

The async function to execute during post-processing.

Returns

[AsyncFuncPostProcessor](#)

EnsureComponent<T>()

Creates a post-processor that will ensure the loaded avatar has a component of type **T**.

```
public static ActionPostProcessor EnsureComponent<T>() where T : Component
```

Returns

[ActionPostProcessor](#)

Type Parameters

T

The component type.

EnsureDependentComponent<T>()

Creates a post-processor that ensures the avatar has a component of type **T** and attaches it as dependent on the loaded avatar.

```
public static ActionPostProcessor EnsureDependentComponent<T>() where T :  
Component, IAvatarDependentComponent
```

Returns

[ActionPostProcessor](#)

Type Parameters

T

The dependent component type.

OverrideMaterials(MaterialOverrideConfig)

Creates a post-processor that recreates avatar materials using alternative shaders and applies configured property and keyword overrides.

```
public static MaterialOverridePostProcessor OverrideMaterials(MaterialOverrideConfig config)
```

Parameters

config [MaterialOverrideConfig](#)

Configuration that defines material override rules, including source shaders, target shaders, and property and keyword mappings.

Returns

[MaterialOverridePostProcessor](#)

SwapShaders(ShaderSwapConfig)

Creates a post-processor that swaps shaders of the loaded avatar based on the given **config**.

```
public static ShaderSwapPostProcessor SwapShaders(ShaderSwapConfig config)
```

Parameters

config [ShaderSwapConfig](#)

Configuration that includes a map of shaders to be replaced and their replacements.

Returns

[ShaderSwapPostProcessor](#)

When(Func<LoadedAv, AvSourceData, bool>, IAsyncAvPostProcessor)

Creates a conditional asynchronous post-processor that runs only when the given condition is met.

```
public static ConditionalPostProcessor When(Func<LoadedAv, AvSourceData, bool> condition,
IAsyncAvPostProcessor onCondition)
```

Parameters

condition Func<[LoadedAv](#), [AvSourceData](#), bool>

A predicate determining whether the post-processor should run.

onCondition [IAsyncAvPostProcessor](#)

The async post-processor to execute when the condition is met.

Returns

[ConditionalPostProcessor](#)

When(Func<LoadedAv, AvSourceData, bool>, IAvPostProcessor)

Creates a conditional post-processor that runs only when the given condition is met.

```
public static ConditionalPostProcessor When(Func<LoadedAv, AvSourceData, bool> condition,
IAvPostProcessor onCondition)
```

Parameters

condition Func<[LoadedAv](#), [AvSourceData](#), bool>

A predicate determining whether the post-processor should run.

onCondition [IAvPostProcessor](#)

The post-processor to execute when the condition is met.

Returns

[ConditionalPostProcessor](#)

WhenImportedWith<T>(IAsyncAvPostProcessor)

Creates a conditional asynchronous post-processor that runs only when the avatar is imported using an importer of type **T**.

```
public static ConditionalPostProcessor WhenImportedWith<T>(IAsyncAvPostProcessor  
onCondition) where T : IAvImporter
```

Parameters

onCondition [IAsyncAvPostProcessor](#)

The post-processor to execute when the condition is met.

Returns

[ConditionalPostProcessor](#)

Type Parameters

T

The importer type that triggers execution of **onCondition**.

WhenImportedWith<T>(IAVPostProcessor)

Creates a conditional post-processor that runs only when the avatar is imported using an importer of type **T**.

```
public static ConditionalPostProcessor WhenImportedWith<T>(IAVPostProcessor onCondition)  
where T : IAvImporter
```

Parameters

onCondition [IAVPostProcessor](#)

The post-processor to execute when the condition is met.

Returns

[ConditionalPostProcessor](#)

Type Parameters

T

The importer type that triggers execution of **onCondition**.

WhenLoadedWith<T>(IAsyncAvPostProcessor)

Creates a conditional asynchronous post-processor that runs only when the avatar data is loaded using a loader of type **T**.

```
public static ConditionalPostProcessor WhenLoadedWith<T>(IAsyncAvPostProcessor onCondition)
where T : IAvDataLoader
```

Parameters

onCondition [IAsyncAvPostProcessor](#)

The post-processor to execute when the condition is met.

Returns

[ConditionalPostProcessor](#)

Type Parameters

T

The loader type that triggers execution of **onCondition**.

WhenLoadedWith<T>(IAvPostProcessor)

Creates a conditional post-processor that runs only when the avatar data is loaded using a loader of type **T**.

```
public static ConditionalPostProcessor WhenLoadedWith<T>(IAvPostProcessor onCondition) where
T : IAvDataLoader
```

Parameters

`onCondition` [IAvPostProcessor](#)

The post-processor to execute when the condition is met.

Returns

[ConditionalPostProcessor](#)

Type Parameters

`T`

The loader type that triggers execution of `onCondition`.

Class CachePostProcessor

Namespace: [Uralstech.AvLoader.PostProcessors](#)

Post-processing step that saves the loaded avatar's data to a local directory for future use. Skips execution if the avatar was originally loaded using [FileAvDataLoader](#), unless overridden via [IgnoreOriginalLoaderType](#).

```
public class CachePostProcessor : IAsyncAvPostProcessor
```

Inheritance

object ← CachePostProcessor

Implements

[IAsyncAvPostProcessor](#)

Remarks

This only works for formats where the model does not depend on resources stored separately from the source file ([Model](#)). For example, avatars loaded from OBJ files with external texture files will not work.

Constructors

CachePostProcessor(string, bool)

```
public CachePostProcessor(string baseDirectory, bool useAvatarIdAsChildDirName = true)
```

Parameters

baseDirectory string

useAvatarIdAsChildDirName bool

Fields

BaseDirectory

The base directory to save the avatar to.

```
public string BaseDirectory
```

Field Value

string

Remarks

If [UseAvatarIdAsDirName](#) is [true](#) (default), the avatar will be stored in a new directory at "[BaseDirectory/Id](#)".

BustRenderJPEGQuality

The quality to encode the avatar's bust render, if saving as JPEG.

```
public int BustRenderJPEGQuality
```

Field Value

int

FullRenderJPEGQuality

The quality to encode the avatar's full render, if saving as JPEG.

```
public int FullRenderJPEGQuality
```

Field Value

int

IgnoreOriginalLoaderType

Should the avatar be saved even if it was loaded from the local filesystem?

```
public bool IgnoreOriginalLoaderType
```

Field Value

bool

UseAvatarIdAsDirName

If [true](#) (default), the avatar will be stored in a new directory at "[BaseDirectory](#)/[Id](#)". Otherwise, [BaseDirectory](#) is used as-is.

```
public bool UseAvatarIdAsDirName
```

Field Value

bool

Properties

BustRenderImageFormat

The file extension and format for saving the avatar's bust render. Cannot be presumptive, like [JPEGAny](#).

```
public AvImageFileExtension BustRenderImageFormat { get; set; }
```

Property Value

[AvImageFileExtension](#)

FullRenderImageFormat

The file extension and format for saving the avatar's full render. Cannot be presumptive, like [JPEGAny](#).

```
public AvImageFileExtension FullRenderImageFormat { get; set; }
```

Property Value

[AvImageFileExtension](#)

Methods

PostProcessAsync(LoadedAv, AvSourceData, CancellationToken)

Runs a post-processing step on the given avatar.

```
public Awaitable PostProcessAsync(LoadedAv avatar, AvSourceData rawData, CancellationTokentoken = default)
```

Parameters

avatar [LoadedAv](#)

The loaded avatar to process.

rawData [AvSourceData](#)

The raw data of the loaded avatar.

token [CancellationToken](#)

Returns

[Awaitable](#)

Class ConditionalPostProcessor

Namespace: [Uralstech.AvLoader.PostProcessors](#)

Post-processing step that runs based on the given condition.

```
public class ConditionalPostProcessor : IAvPostProcessor, IAsyncAvPostProcessor
```

Inheritance

object ← ConditionalPostProcessor

Implements

[IAvPostProcessor](#), [IAsyncAvPostProcessor](#)

Constructors

ConditionalPostProcessor(Func<LoadedAv, AvSourceData, bool>, IAsyncAvPostProcessor)

```
public ConditionalPostProcessor(Func<LoadedAv, AvSourceData, bool> condition,  
    IAsyncAvPostProcessor asyncPostProcessor)
```

Parameters

condition Func<[LoadedAv](#), [AvSourceData](#), bool>

asyncPostProcessor [IAsyncAvPostProcessor](#)

ConditionalPostProcessor(Func<LoadedAv, AvSourceData, bool>, IAvPostProcessor)

```
public ConditionalPostProcessor(Func<LoadedAv, AvSourceData, bool> condition,  
    IAvPostProcessor postProcessor)
```

Parameters

condition Func<[LoadedAv](#), [AvSourceData](#), bool>

postProcessor [IAvPostProcessor](#)

Fields

Condition

Condition callback.

```
public Func<LoadedAv, AvSourceData, bool> Condition
```

Field Value

Func<[LoadedAv](#), [AvSourceData](#), bool>

Properties

PostProcessor

Post-processor to run if [Condition](#) is met, can be of any type inheriting from [IAvPostProcessor](#) or [IAsyncAvPostProcessor](#).

```
public object PostProcessor { get; set; }
```

Property Value

object

Methods

PostProcess(LoadedAv, AvSourceData)

Runs a post-processing step on the given avatar.

```
public void PostProcess(LoadedAv avatar, AvSourceData rawData)
```


Parameters

avatar [LoadedAv](#)

The loaded avatar to process.

rawData [AvSourceData](#)

The raw data of the loaded avatar.

PostProcessAsync(LoadedAv, AvSourceData, CancellationToken)

Runs a post-processing step on the given avatar.

```
public Awaitable PostProcessAsync(LoadedAv avatar, AvSourceData rawData, CancellationToken token = default)
```

Parameters

avatar [LoadedAv](#)

The loaded avatar to process.

rawData [AvSourceData](#)

The raw data of the loaded avatar.

token [CancellationToken](#)

Returns

[Awaitable](#)

Interface IAvatarDependentComponent

Namespace: [Uralstech.AvLoader.PostProcessors](#)

Marks a Component as semantically dependent on a [LoadedAv](#) instance.

```
public interface IAvatarDependentComponent
```

Methods

OnAvatarAttached(LoadedAv)

Called when the component is attached to an avatar via a dependent-component post-processor.

```
void OnAvatarAttached(LoadedAv avatar)
```

Parameters

avatar [LoadedAv](#)

The loaded avatar this component depends on.

Class MaterialOverrideConfig

Namespace: [Uralstech.AvLoader.PostProcessors](#)

Configuration data for [MaterialOverridePostProcessor](#).

```
public class MaterialOverrideConfig : ScriptableObject, ISerializationCallbackReceiver
```

Inheritance

object ← MaterialOverrideConfig

Implements

ISerializationCallbackReceiver

Fields

MaterialOverrides

Runtime map of source shaders to their material override definitions.

```
public Dictionary<Shader, RuntimeMaterialOverrideDefinition> MaterialOverrides
```

Field Value

Dictionary<Shader, [RuntimeMaterialOverrideDefinition](#)>

Class MaterialOverridePostProcessor

Namespace: [Uralstech.AvLoader.PostProcessors](#)

Post-processing step that recreates avatar materials using alternative shaders and copies selected property and keyword values from the original materials.

```
public class MaterialOverridePostProcessor : IAvPostProcessor
```

Inheritance

object ← MaterialOverridePostProcessor

Implements

[IAvPostProcessor](#)

Constructors

MaterialOverridePostProcessor(MaterialOverrideConfig)

```
public MaterialOverridePostProcessor(MaterialOverrideConfig configuration)
```

Parameters

configuration [MaterialOverrideConfig](#)

Fields

Configuration

Configuration which defines the material override rules.

```
public MaterialOverrideConfig Configuration
```

Field Value

[MaterialOverrideConfig](#)

Methods

PostProcess(LoadedAv, AvSourceData)

Runs a post-processing step on the given avatar.

```
public void PostProcess(LoadedAv avatar, AvSourceData rawData)
```

Parameters

avatar [LoadedAv](#)

The loaded avatar to process.

rawData [AvSourceData](#)

The raw data of the loaded avatar.

Class ShaderSwapConfig

Namespace: [Uralstech.AvLoader.PostProcessors](#)

Configuration data for [ShaderSwapPostProcessor](#).

```
public class ShaderSwapConfig : ScriptableObject, ISerializationCallbackReceiver
```

Inheritance

object ← ShaderSwapConfig

Implements

ISerializationCallbackReceiver

Fields

FallbackShader

Optional shader used when no entry exists in [ShaderMap](#).

```
public Shader? FallbackShader
```

Field Value

Shader?

ShaderMap

Maps source shaders to their replacement shaders.

```
public Dictionary<Shader, Shader> ShaderMap
```

Field Value

Dictionary<Shader, Shader>

Class ShaderSwapPostProcessor

Namespace: [Uralstech.AvLoader.PostProcessors](#)

Post-processing step that swaps the avatar's shaders with the provided set of shaders.

```
public class ShaderSwapPostProcessor : IAvPostProcessor
```

Inheritance

object ← ShaderSwapPostProcessor

Implements

[IAvPostProcessor](#)

Constructors

ShaderSwapPostProcessor(ShaderSwapConfig)

```
public ShaderSwapPostProcessor(ShaderSwapConfig configuration)
```

Parameters

configuration [ShaderSwapConfig](#)

Fields

Configuration

Configuration which defines the shaders to swap.

```
public ShaderSwapConfig Configuration
```

Field Value

[ShaderSwapConfig](#)

Methods

PostProcess(LoadedAv, AvSourceData)

Runs a post-processing step on the given avatar.

```
public void PostProcess(LoadedAv avatar, AvSourceData _)
```

Parameters

avatar [LoadedAv](#)

The loaded avatar to process.

_ [AvSourceData](#)

Namespace Uralstech.AvLoader.PostProcessors. Rendering

Classes

[RuntimeMaterialOverrideDefinition](#)

Defines how a material using a specific shader is recreated using another shader, with selected properties and keywords copied.

[ShaderKeywordMapping](#)

Describes how a single shader keyword is copied from a source material to a target material.

[ShaderKeywordRule](#)

Defines a conditional rule that enables or disables a target shader keyword based on the state of a source keyword.

[ShaderMapping](#)

Defines a mapping from a source shader to a target shader.

[ShaderPropertyMapping](#)

Describes how a single shader property value is copied from a source material to a target material.

Enums

[ShaderPropertyType](#)

Represents the supported shader property data types that can be transferred between materials, based on the Material `.GetX` and `.SetX` methods.

Class RuntimeMaterialOverrideDefinition

Namespace: [Uralstech.AvLoader.PostProcessors.Rendering](#)

Defines how a material using a specific shader is recreated using another shader, with selected properties and keywords copied.

```
public class RuntimeMaterialOverrideDefinition
```

Inheritance

object ← RuntimeMaterialOverrideDefinition

Constructors

RuntimeMaterialOverrideDefinition(Shader, IReadOnlyList<ShaderPropertyMapping>, IReadOnlyList<ShaderKeywordMapping>?, IReadOnlyList<ShaderKeywordRule>?)

```
public RuntimeMaterialOverrideDefinition(Shader target, IReadOnlyList<ShaderPropertyMapping>
propertyMappings, IReadOnlyList<ShaderKeywordMapping>? keywordMappings = null,
IReadOnlyList<ShaderKeywordRule>? keywordRules = null)
```

Parameters

target Shader

propertyMappings IReadOnlyList<[ShaderPropertyMapping](#)>

keywordMappings IReadOnlyList<[ShaderKeywordMapping](#)>

keywordRules IReadOnlyList<[ShaderKeywordRule](#)>

Fields

Target

The shader used by the newly created material.

```
public Shader Target
```

Field Value

Shader

Properties

KeywordMappings

Optional keyword mappings that directly copy enabled/disabled states from source keywords to target keywords.

```
public IReadOnlyList<ShaderKeywordMapping>? KeywordMappings { get; set; }
```

Property Value

IReadOnlyList<[ShaderKeywordMapping](#)>

Remarks

For conditional or overridden behavior, use [KeywordRules](#) instead.

Exceptions

ArgumentException

Thrown if any mapping is invalid or if duplicate source or target keywords are found.

KeywordRules

Optional rules that conditionally enable or disable shader keywords on the new material.

```
public IReadOnlyList<ShaderKeywordRule>? KeywordRules { get; set; }
```

Property Value

`ReadOnlyList<ShaderKeywordRule>`

Exceptions

ArgumentException

Thrown if any rule has an empty or null source or target keyword.

PropertyMappings

Property mappings used to copy values from the source material to the newly created material.

```
public IReadOnlyList<ShaderPropertyMapping> PropertyMappings { get; set; }
```

Property Value

`ReadOnlyList<ShaderPropertyMapping>`

Exceptions

ArgumentException

Thrown if any mapping is invalid or if duplicate source or target properties are found.

Class ShaderKeywordMapping

Namespace: [Uralstech.AvLoader.PostProcessors.Rendering](#)

Describes how a single shader keyword is copied from a source material to a target material.

```
[Serializable]  
public class ShaderKeywordMapping
```

Inheritance

object ← ShaderKeywordMapping

Remarks

Use [ShaderKeywordRule](#) for conditional behavior. A mapping directly mirrors the enabled state of [Source](#) onto [Target](#).

Fields

Source

The name of the keyword on the source shader.

```
public string? Source
```

Field Value

string

Target

The name of the corresponding keyword on the target shader.

```
public string? Target
```

Field Value

string

Methods

IsValid()

Returns [true](#) if both source and target keyword names are non-empty.

```
public bool IsValid()
```

Returns

bool

IsValid(IReadOnlyList<ShaderKeywordMapping>?)

Returns [true](#) if all keyword mappings in the list are valid and contain no duplicates.

```
public static bool IsValid(IReadOnlyList<ShaderKeywordMapping>? mappings)
```

Parameters

mappings IReadOnlyList<[ShaderKeywordMapping](#)>

Returns

bool

Class ShaderKeywordRule

Namespace: [Uralstech.AvLoader.PostProcessors.Rendering](#)

Defines a conditional rule that enables or disables a target shader keyword based on the state of a source keyword.

```
[Serializable]  
public class ShaderKeywordRule
```

Inheritance

object ← ShaderKeywordRule

Remarks

If you want to copy the enabled/disabled state of a keyword to another one, use [ShaderKeyword Mapping](#) instead.

Fields

Source

The source shader keyword to test.

```
public string? Source
```

Field Value

string

SourceRequiredState

The required enabled state of the source keyword for this rule to apply.

```
public bool SourceRequiredState
```

Field Value

bool

Target

The target shader keyword to modify.

```
public string? Target
```

Field Value

string

TargetResultState

The keyword state applied to the target material when the rule matches.

```
public bool TargetResultState
```

Field Value

bool

Methods

IsValid()

Returns [true](#) if both source and target keyword names are non-empty.

```
public bool IsValid()
```

Returns

bool

IsValid(IReadOnlyList<ShaderKeywordRule> ?)

Returns [true](#) if all keyword rules in the list are valid.

```
public static bool IsValid(IReadOnlyList<ShaderKeywordRule>? rules)
```

Parameters

rules IReadOnlyList<[ShaderKeywordRule](#)>

Returns

bool

Class ShaderMapping

Namespace: [Uralstech.AvLoader.PostProcessors.Rendering](#)

Defines a mapping from a source shader to a target shader.

```
[Serializable]  
public class ShaderMapping
```

Inheritance

object ← ShaderMapping

Fields

Source

The original shader to be replaced.

```
public Shader? Source
```

Field Value

Shader?

Target

The shader that replaces the source shader.

```
public Shader? Target
```

Field Value

Shader?

Methods

IsValid()

Returns [true](#) if both source and target shaders are assigned.

```
public bool IsValid()
```

Returns

bool

Class ShaderPropertyMapping

Namespace: [Uralstech.AvLoader.PostProcessors.Rendering](#)

Describes how a single shader property value is copied from a source material to a target material.

```
[Serializable]  
public class ShaderPropertyMapping
```

Inheritance

object ← ShaderPropertyMapping

Fields

Source

The name of the property on the source shader.

```
public string? Source
```

Field Value

string

Target

The name of the corresponding property on the target shader.

```
public string? Target
```

Field Value

string

Type

The expected data type of the shader property.

```
public ShaderPropertyType Type
```

Field Value

[ShaderPropertyType](#)

Methods

IsValid()

Returns [true](#) if both source and target property names are non-empty.

```
public bool IsValid()
```

Returns

bool

Enum ShaderPropertyType

Namespace: [Uralstech.AvLoader.PostProcessors.Rendering](#)

Represents the supported shader property data types that can be transferred between materials, based on the Material [.GetX](#) and [.SetX](#) methods.

```
public enum ShaderPropertyType
```

Fields

[Color](#) = 2

[Float](#) = 0

[Inverse01RangeFloat](#) = 5

Inverts the float value in 0-1 range before assignment.

E.g. assigning a "smoothness" factor to a shader that expects a "roughness" factor.

[Matrix](#) = 4

[Texture](#) = 3

[Vector](#) = 1

Namespace Uralstech.AvLoader.Utils

Classes

[ListExtensions](#)

Utility extensions for lists.

[TextureExtensions](#)

Utility extensions for texture-related operations.

[UnityWebRequestException](#)

Exception thrown when a UnityWebRequest fails, providing detailed request information.

[WebRequestExtensions](#)

Utility extensions for web requests.

Class ListExtensions

Namespace: [Uralstech.AvLoader.Utils](#)

Utility extensions for lists.

```
public static class ListExtensions
```

Inheritance

object ← ListExtensions

Remarks

This class is [public](#) to allow package users to reuse these extensions if useful. However, it should not be considered stable and is not part of the supported public API. It exists solely as an internal utility and may change or be removed at any time.

Methods

CreateCopy<T>(IReadOnlyList<T>)

Creates a shallow copy of the current list.

```
public static T[] CreateCopy<T>(this IReadOnlyList<T> current)
```

Parameters

current IReadOnlyList<T>

Returns

T[]

Type Parameters

T

Class TextureExtensions

Namespace: [Uralstech.AvLoader.Utils](#)

Utility extensions for texture-related operations.

```
public static class TextureExtensions
```

Inheritance

object ← TextureExtensions

Remarks

This class is [public](#) to allow package users to reuse these extensions if useful. However, it should not be considered stable and is not part of the supported public API. It exists solely as an internal utility and may change or be removed at any time.

Methods

TryDecodeImage(byte[]?, out Texture2D?, string?, bool)

```
public static bool TryDecodeImage(this byte[]? current, out Texture2D? result, string? name  
= null, bool throwOnFail = false)
```

Parameters

current byte[]

result Texture2D?

name string

throwOnFail bool

Returns

bool

TryDecodeImage(ReadOnlySpan<byte>, out Texture2D?, string?, bool)

Tries decoding a Texture2D from binary data using ImageConversion.LoadImage(Texture2D, ReadOnlySpan<byte>).

```
public static bool TryDecodeImage(this ReadOnlySpan<byte> current, out Texture2D? result,
    string? name = null, bool throwOnFail = false)
```

Parameters

current ReadOnlySpan<byte>

result Texture2D?

name string

Name to give the created texture, and for error logs.

throwOnFail bool

Should the method throw if it couldn't load the image data?

Returns

bool

TryGetTextureAsync(UnityWebRequest?, bool, string?, CancellationToken)

Awaits for the DownloadHandlerTexture of the current UnityWebRequest to finish processing its texture.

```
public static Awaitable<Texture2D?> TryGetTextureAsync(this UnityWebRequest? request, bool
    throwOnFail = false, string? name = null, CancellationToken token = default)
```

Parameters

request UnityWebRequest?

throwOnFail bool

Should the method throw if it couldn't load the image data?

name string

Name for error logs.

token CancellationToken

Returns

Awaitable<Texture2D?>

Class UnityWebRequestException

Namespace: [Uralstech.AvLoader.Utils](#)

Exception thrown when a UnityWebRequest fails, providing detailed request information.

```
public sealed class UnityWebRequestException : IOException
```

Inheritance

object ← Exception ← SystemException ← IOException ← UnityWebRequestException

Constructors

UnityWebRequestException(UnityWebRequest)

```
public UnityWebRequestException(UnityWebRequest request)
```

Parameters

request UnityWebRequest

Fields

DownloadHandlerError

The error message reported by the DownloadHandler.

```
public readonly string DownloadHandlerError
```

Field Value

string

Error

The error message reported by the UnityWebRequest.

```
public readonly string Error
```

Field Value

string

ResponseCode

The HTTP response code (if any).

```
public readonly long ResponseCode
```

Field Value

long

Url

The URL that failed to load.

```
public readonly string Url
```

Field Value

string

Class WebRequestExtensions

Namespace: [Uralstech.AvLoader.Utls](#)

Utility extensions for web requests.

```
public static class WebRequestExtensions
```

Inheritance

object ← WebRequestExtensions

Remarks

This class is [public](#) to allow package users to reuse these extensions if useful. However, it should not be considered stable and is not part of the supported public API. It exists solely as an internal utility and may change or be removed at any time.

Methods

AbortAll(List<UnityWebRequest>)

Aborts all UnityWebRequests in the current list.

```
public static void AbortAll(this List<UnityWebRequest> current)
```

Parameters

current List<UnityWebRequest>

AsTask(AsyncOperation)

Wraps an AsyncOperation as a System.Threading.Tasks.Task.

```
public static Task AsTask(this AsyncOperation current)
```

Parameters

current AsyncOperation

Returns

Task

DisposeAll(List<UnityWebRequest>)

Disposes all UnityWebRequests in the current list.

```
public static void DisposeAll(this List<UnityWebRequest> current)
```

Parameters

current List<UnityWebRequest>

GetErred(List<UnityWebRequest>)

Enumerates the erred/failed UnityWebRequests in the current list.

```
public static IEnumerable<UnityWebRequest> GetErred(this List<UnityWebRequest> current)
```

Parameters

current List<UnityWebRequest>

Returns

IEnumerable<UnityWebRequest>

SendAll(List<UnityWebRequest>)

Sends all UnityWebRequests in the current list, and creates a System.Threading.Tasks.Task[] for their async operations.

```
public static Task[] SendAll(this List<UnityWebRequest> current)
```

Parameters

current List<UnityWebRequest>

Returns

Task[]